AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM340
Natural Language Processing

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 9 / April /2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **two** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

Multiple Choice Questions **(10 marks)**

Question 1, (10 marks, 1 mark for each question)
Answer the following MCQ's by selecting **the best** answer for each question. Write your **answer letter** in the answer sheet.

1. ..... is considered a pre-processing step in the NLP pipeline?
   a) Model evaluation
   b) Tokenization
   c) Deployment
   d) Training
2. Which ambiguity involves the meaning of a word or sentence?
   a) Phonological ambiguity
   b) Morphological ambiguity
   c) Semantic ambiguity
   d) Structural ambiguity
3. Which approach uses minimal training data to generalize across tasks?
   a) Rule-based NLP
   b) Few-shot Learning
   c) Bayesian Methods
   d) Statistical NLP
4. What does the Kleene star (*) operator do in regular expressions?
   a) Matches 1 or more occurrences of the preceding character
   b) Matches exactly two occurrences of the preceding character
   c) Matches 0 or more occurrences of the preceding character
   d) Matches a space
5. Which regular expression pattern matches any digit?
   a) \w
   b) \d
   c) \s
   d) \n
6. Which model assumes that words are independent of each other?
   a) Unigram model
   b) Bigram model
   c) Trigram model
   d) 5-gram model
7. How is the perplexity of a language model related to its performance?
   a) Lower perplexity indicates better performance
   b) Higher perplexity indicates better performance
   c) Perplexity does not affect performance
   d) Perplexity should be as high as possible
8. Which of the following does Byte-Pair Encoding (BPE) handle in tokenization?
   a) Splitting sentences into words
   b) Merging frequent pairs of characters
   c) Counting words in a sentence
   d) Removing stop words
9. What assumption does the Markov model make in N-gram language models?
   a) Future words depend only on the previous N words
   b) Future words depend on the entire sentence
   c) All words are independent
   d) All sentences are independent
10. What is meant by "Bag of Words" representation?
    a) The order of words is crucial
    b) Words are represented by their frequency without considering the order
    c) Words are grouped based on meaning
    d) The model focuses on stop words

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PART 2

Short Essay Questions **(10 marks)**

Question 2, (10 marks)
Answer the following questions:

a) Compare generative and discriminative classifiers, providing an example of each? [4 marks]
b) List and describe the three main approaches to do NLP? [3 marks]
c) Explain the role of Laplace smoothing in Naive Bayes classification? [3 marks]

PART 3

Short Problems/Questions **(40 marks)**

Question 3, (10 marks)
Calculate the Minimum Edit Distance for transforming "**flaw**" into "**lawn**" using uniform cost operations (cost = 1 for insertion, deletion, substitution):

a) Complete the distance matrix for the given words. [8 marks]

4	w					
3	a					
2	l					
1	f					
0	#					
		#	l	a	w	n
		0	1	2	3	4

b) Using the distance matrix, what is the minimum edit distance value. [2 marks]

Question 4, (10 marks)
Suppose that you have a language L, given a sentence **S** = "**I want coffee**", with the following bigram probabilities:

P(I) = 0.4 **P(want | I) = 0.5** **P(coffee | want) = 0.4**

a) Compute the probability of **S** for this model. [4 marks]
b) Explain the concept of **perplexity**. [2 marks]
c) Calculate the **perplexity** of the sentence **S** for this model. [4 marks]

Question 5, (10 marks)
Suppose that you have a text classifier and test it on a dataset containing 150 examples. And the confusion matrix was:

		True Class	
		Positive	Negative
Predicted Class	Positive	45	15
	Negative	30	60

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a) According to the confusion matrix, compute the **accuracy** value? [5 marks]
b) According to the confusion matrix, compute the **precision** value? [5 marks]

Question 6, (10 marks)
Assume the following likelihoods for each word being part of a ***Spam vs. Ham*** email classification, and **equal prior probabilities** for each class:

Word	Spam	Ham
offer	0.20	0.05
free	0.15	0.03
check	0.04	0.10
out	0.06	0.12
latest	0.10	0.02

And given the document d = "offer free latest check out", Consider the ***naive Bayes classifier***:

a) Compute $p(d|Spam)$? [4 marks]
b) Compute $p(d|Ham)$? [4 marks]
c) What class will Naive Bayes assign to the document d? [2 marks]

End of questions

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